

AI in Genomics 2026

Worksheet · Practicals 3 and 4

Tunisia · 14–18 September 2026 · Yu Sugihara, The Sainsbury Laboratory

Set up before we start

Setup guide https://andyposbe.github.io/AI-in-Genomics_Workshop_Tunisia_2026/setup.html

Data, Practical 3 https://github.com/andyposbe/AI-in-Genomics_Workshop_Tunisia_2026/tree/main/data/practical-03_compare

Data, Practical 4 https://github.com/andyposbe/AI-in-Genomics_Workshop_Tunisia_2026/tree/main/data/practical-04_design

ChimeraX <https://www.cgl.ucsf.edu/chimerax/download.html>

Protenix account <https://protenix-server.com/login>

PRACTICAL 3 · DAY 2, 09:30–11:15

Compare

Align structures to reveal hidden relationships beyond sequence similarity

Exercise 1 · Comparing sequences

Dataset 1 — *dataset-1/Proteins_1-4.fasta*, containing Protein_1 to Protein_4

COPY THE FOUR SEQUENCES FROM HERE

```
>Protein_1
DLATGDLHHSAVCVTKRVNFFGHKTFQISPNATMCACTFYKGRNTGTNPWDKCEDCKYDGLKCTSENGTIGGDEFKYCEKKCGAKGSEAD
>Protein_2
GIEEGNYCCAGACISSVTVVNGVVSYTKNIAGTKCACELMRALDSGSAKGDAPNAYFNGENCLDPDCNLSYSRLRHFCVNECGCAGAICW
>Protein_3
KRAIDLSRERDPNFFDHPGIPVPECFWFMFKNVNRQDAGTCYSSWKMDMKVGPNNVHIKSDNCNLSGDFPPGWIVLGKKRPGF
>Protein_4
RNCRIWQDMGSMVWQEVVVVTPPVTVDIITKRHGAFSLFVPVGC GIRD TG GALRATETDDPW
```

Select from “>Protein_1” to the end of the last line, then paste.

1.1 Search the NCBI database

NCBI blastp <https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins>

Paste all four sequences at once, in FASTA format ①. Leave “Align two or more sequences” unchecked ②, keep the non-redundant protein database (nr) ③, and run the search ④.

BLAST® » blastp suite

Standard Protein BLAST

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) [Query subrange](#)

From

To

Or, upload file [Browse...](#) No file selected. [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#) **leave this unchecked**

Choose Search Set

Database **Non-redundant protein sequences (nr)** [?](#) **choose the database**

Organism [Optional](#) [Add organism](#)

Exclude [Optional](#) ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☒ Quick BLASTP (Accelerated protein-protein BLAST)

☐ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

[Choose a BLAST algorithm](#) [?](#)

BLAST Search database nr using Blastp (protein-protein BLAST)

☐ Show results in a new window

Q1.1a How many hits does each sequence return?

Q1.1b What are the hit proteins called?

1.2 Search your own sequences

The same page, with “Align two or more sequences” ticked ①. Now the database is your own data: your sequences as the query ②, your sequences as the subject ③, then BLAST ④.

BLAST® » blastp suite

Align Sequences Protein BLAST

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) [Query subrange](#)

From

To

Or, upload file **2** [Browse...](#) No file selected. [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

1 ☒ Align two or more sequences [?](#)

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) [Subject subrange](#)

From

To

Or, upload file **3** [Browse...](#) No file selected. [?](#)

Program Selection

Algorithm

☒ blastp (protein-protein BLAST)

[Choose a BLAST algorithm](#) [?](#)

4 **BLAST** Search protein sequence using Blastp (protein-protein BLAST)

☒ Show results in a new window

Q1.2 Which of your sequences find each other, and how strong is the alignment?

1.3 Domain annotation with InterProScan

InterProScan <https://www.ebi.ac.uk/interpro/search/sequence/>

Q1.3a Which proteins carry an annotated domain?

Q1.3b Open one of the domain entries. Besides the name, what else does InterPro show you — which taxa carry it, which structures are known, which other domains it appears with?

Exercise 2 · Comparing protein structures

Dataset 2 — all four files in dataset-2: Protein_1_BAS4.pdb, Protein_2_IF.pdb, Protein_3_AVR-PikD.pdb, Protein_4_AVR-Mgk1.pdb

2.1 Superposing in ChimeraX

Open all four PDB files, then align each structure onto the first, one at a time:

```
mm #2 to #1
mm #3 to #1
mm #4 to #1
```

Once the syntax is familiar, the same three alignments fit in one command:

```
mm #2-4 to #1
```

Q2.1a Which structures look similar?

Q2.1b What is the difference between two similar structures and two proteins sharing a fold?

Q2.1c Can a superposition alone show that two proteins share a fold?

2.2 Scoring the alignment

TM-align <https://zhanggroup.org/TM-align/>

Two pairs are enough here: Protein 1 with Protein 2 ①②, then Protein 3 with Protein 4.

TM-align on-line (view an example of output)

Note: This server is only for pair-wise structure comparison. If you want to match one protein structure with all proteins in the PDB library, you can do it in CQFACTOR Server.

- Input Structure 1 in **PDB format** or **PDBx/mmCIF format** (mandatory):
Please copy and paste your structure file here. [Sample input](#)

Or upload the structure file:
 No file selected.

- Input Structure 2 in **PDB format** or **PDBx/mmCIF format** (mandatory):
Please copy and paste your structure file here. [Sample input](#)

Or upload the structure file:
 No file selected.

Input Email: (optional, where results will be sent to)

Upload the first structure ①

Upload the second structure ②

- Q2.2a** What TM-score does each of the two pairs give?
- Q2.2b** Two scores come back per pair. Why do they differ?
- Q2.2c** Do the scores suggest that Protein 3 and Protein 4 share a fold?

Exercise 3 · Searching with Foldseek

Dataset 2 — one file only: *dataset-2/Protein_4_AVR-Mgk1.pdb*

Foldseek <https://search.foldseek.com/search>

Foldseek Search

Databases & search settings

Databases

- ☒ AlphaFold/UniProt50 v6
- ☒ AlphaFold/Swiss-Prot v6
- ☒ AlphaFold/Proteome v6
- ☒ CATH50 4.3.0
- ☒ BFD 2023_02_v2
- ☒ MGnify-ESM30 v1
- ☒ PDB100 20240101
- ☒ GMCL 2204
- ☒ BFMD 20240623

Mode

☐ 3Di/AA
☒ TM-align
☐ LoL-align

Taxonomic filter

☐ Iterative search

Click "SEARCH" ③

SEARCH

Summary
 Search all available databases with Foldseek in TM-align mode.

Upload the PDB file ①

Select TM-align ②

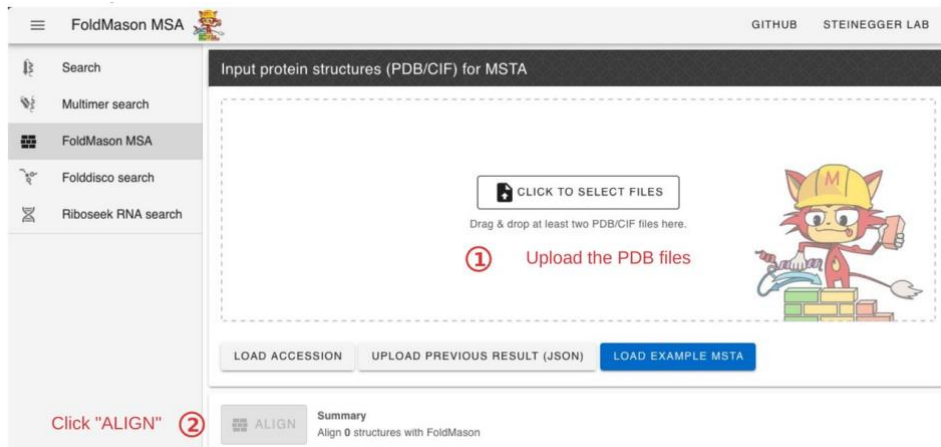
Three alignment modes. 3Di/AA — local similarity from structure and sequence together. TM-align — overall structural similarity. LoL-align — local similarity indicative of homology.

- Q3a** What are the top hits, and did BLAST or InterProScan find them too?
- Q3b** How do the results differ between the databases?

Exercise 4 · Aligning many structures with FoldMason

Dataset 3 — the 80 structures in dataset-3, MAX-fold effectors from Lahfa et al., 2024

FoldMason <https://search.foldseek.com/foldmason>



Q4a In the tree, which structures sit closest to AVR-Mgk1 and AVR-PikD?

Q4b Which residues stay conserved across the family?

PRACTICAL 4 · DAY 2, 13:30–14:45

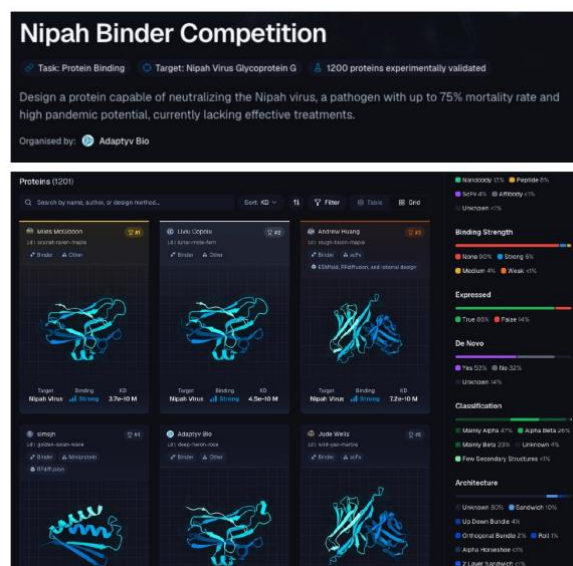
Design

Design de novo structures and binders against a target to create novel solutions

Exercise 5 · What a real competition tells you

Nipah results <https://proteinbase.com/collections/nipah-binder-competition-results>

Competition write-up <https://www.adaptvbio.com/blog/nipah-submissions>

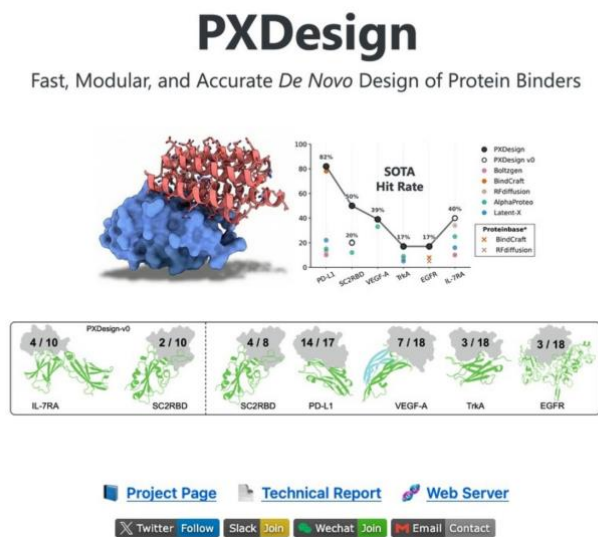


Q5a Which computational score, if any, separates binders from non-binders?

Q5b Why does it matter that the failures are published too?

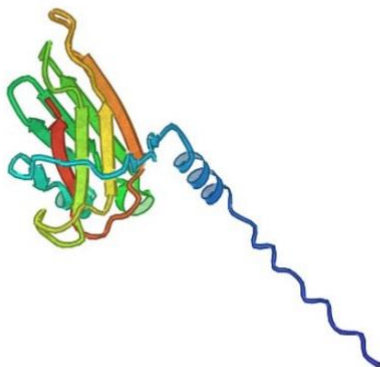
Exercise 6 · Designing with PXDesign

Protenix server <https://protenix-server.com/login>

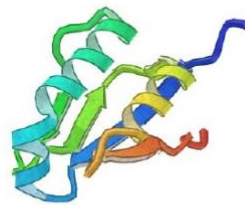


Pick a target: FoaToxA or FoaKP6, or a protein of your own if you have one. Submit a single job in Preview Mode. Extended Mode is slower, but it generates many more candidates, so the chance of a real binder among them is higher.

FoaToxA



FoaKP6



- Q6a** What do you gain by choosing a hotspot?
- Q6b** How difficult is your target on the server's own scale?
- Q6c** Why is every design checked with both AlphaFold2 and Protenix?
- Q6d** What do the check marks beside each design mean?
- Q6e** How many of your designs pass those checks?
- Q6f** Where on the target do your binders sit?

Tools and links

Practical 3 • Compare

Name	Description	URL
NCBI blastp	Sequence similarity search	https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins
InterProScan	Protein domain and family annotation	https://www.ebi.ac.uk/interpro/search/sequence/
TM-align	Pairwise structural alignment, with a TM-score	https://zhanggroup.org/TM-align/
Foldseek	Structure search against whole databases	https://search.foldseek.com/search
FoldMason	Multiple structure alignment and tree	https://search.foldseek.com/foldmason
ProteinMPNN	Inverse folding: designs a sequence that folds into a given structure	https://colab.research.google.com/github/dauparas/ProteinMPNN/blob/main/colab_notebooks/quick_demo.ipynb

Practical 4 • Design

Name	Description	URL
PXDesign	Server for binder generation, filtering and ranking	https://protenix-server.com/login
Adaptyv Bio	Open binder-design competitions	https://proteinbase.com/competitions
Nipah results	Database of designs, scores and measured affinities	https://proteinbase.com/collections/nipah-binder-competition-results
Adaptyv write-up	Blog post analysing the Nipah submissions	https://www.adaptyvbio.com/blog/nipah-submissions
Escalante blog	Blog post on the winning de novo method	https://blog.escalante.bio/winning-the-de-novo-portion-of-the-adaptyv-nipah-binder-competition/
Hallucination vs diffusion	Blog post comparing design strategies	https://blog.escalante.bio/teaching-generative-models-to-hallucinate/